UNITED STATES DISTRICT COURT
EASTERN DISTRICT OF LOUISIANA

KATHLEEN MOLLER,	*	
Plaintiff,	*	
v.	*	**CIVIL ACTION NO. 2:24-CV-228-WBV-DPC**
MARTIAN SALES, INC.; JOPEN, LLC; JOHNSON FOODS, LLC; LP IND., LLC; CAG HOLDINGS, LLC; RMH HOLDINGS, LLC; ABC INSURANCE COMPANY and John Does 1-4,	*	**THE HONORABLE WENDY B. VITTER**
	*	**MAGISTRATE JUDGE DONNA PHILLIPS CURRAULT**
Defendants.	*	

DEFENDANTS' MEMORANDUM SUPPORT OF MOTION TO EXCLUDE PLAINTIFF'S EXPERT JOSEF G. THUNDIYIL UNDER FED. RULE OF EVID. 702

Defendant JOpen, LLC ("Defendant" or "JOpen") hereby files this Motion to Exclude Plaintiff's Expert Witness, Dr. Josef G. Thundiyil, under Federal Rule of Evidence 702 and *Daubert v. Merrell Dow Pharmaceuticals, Inc.*, 509 U.S. 579 (1993) and incorporated memorandum of law.

I. INTRODUCTION

Plaintiff has designated Dr. Thundiyil to offer causation opinions that: (1) "Harmony Moller's cause of death was an adverse cardiorespiratory event caused by kratom" and (2) "Harmony Moller did not cause her own death." Ex. 1, Nov. 18, 2025 Expert Report of Dr. Josef Thundiyil, at 8, 11.

This is a toxic-tort case. Under controlling Fifth Circuit law, Plaintiff cannot present a causation expert to the jury unless that expert first establishes reliable general causation—that kratom is capable of causing the type of fatal event alleged here at the levels present in this case. Only then may the expert attempt to prove specific causation.

Dr. Thundiyil cannot clear that first gate. He concedes that "we don't know the exact level

[of kratom] that causes toxicity," and that it is "nearly impossible … with the existing science, to have a dose that we know causes harm." Ex. 2, Excerpts from Thundiyil Dep. 146:8–9, 147:3–5, Jan. 16, 2026. He admits he cannot identify a fatal dose range, cannot reliably extrapolate from post-mortem levels to antemortem exposure, and cannot determine whether the death sequence was cardiac-first or respiratory-first. Those admissions are not minor weaknesses. They eliminate the foundational element required by the Fifth Circuit.

Despite these concessions, Dr. Thundiyil proposes to tell the jury that the mitragynine concentration in this case was "sufficient to cause adverse events" and that kratom caused Harmony Moller's death. Ex. 1, at 10. That is the precise analytical leap the Fifth Circuit forbids: an inference that bridges an admitted dose-response void with nothing more than conclusion.

The methodological defect deepens at the specific-causation step. Dr. Thundiyil purports to conduct a differential diagnosis, but he applies a quantitative standard to exclude alternative causes while admitting no such scientifically established threshold exists for kratom. He rejects THC, diphenhydramine, and promethazine because they were not present at a "fatal or toxic" level, yet cannot identify any fatal or toxic level for kratom itself. *Id.* That is not differential diagnosis; it is a one-directional inference that treats kratom as sufficient without reliable science while dismissing alternatives under a standard he cannot consistently apply.

His invocation of "Bradford Hill" does not cure these defects. He admits the analysis was only performed "mentally," without a step-by-step application, documentation, or testable framework. Rule 702 does not permit experts to substitute an unarticulated mental checklist for reproducible methodology. Courts evaluate what an expert actually did, not what he calls it.

Under Rule 702 and binding Fifth Circuit precedent, this Court must ensure reliability "at each and every step." *Knight v. Kirby Inland Marine, Inc.*, 482 F.3d 347, 355 (5th Cir. 2007).

Because Dr. Thundiyil cannot establish general causation, cannot reliably apply specific causation, and cannot consistently rule out alternatives under a defensible scientific framework, his causation opinions are inadmissible and must be excluded in full.

II. FACTUAL BACKGROUND

This is a products liability case in which Plaintiff alleges that O.P.M.S. Silver kratom caused Harmony Moller's death on February 6, 2023. *See* ECF. No. 100, at ¶¶ 30, 33. Specifically, Plaintiff claims that Defendants designed an inherently and unreasonably dangerous product, that Defendants failed to adequately warn of the dangers associated with O.P.M.S. Silver kratom, and that Defendants failed to provide adequate instructions on how to avoid such risks. *See* ECF No. 100, at ¶ 61. It is undisputed that the level of mitragynine in Harmony's blood after death was very low: 150 ng/mL. Ex. 1, at 10.

Dr. Thundiyil is a licensed medical doctor practicing in the state of Florida, with Board Certifications in Emergency Medicine, Medical Toxicology, and Occupation and Environmental Medicine. Ex. 1, at 1. He presents himself "as an expert in both acute management of medical conditions (Emergency Medicine) and the effects of chemicals and substances on the human health (Medical Toxicology)." *Id.* Dr. Thundiyil was retained by Plaintiff to provide expert opinions on Harmony Moller's "death and the cause of her death, including the impact of and causative nature of Defendants' [] kratom products." *Id.*

During Dr. Thundiyil's deposition, counsel for Defendant's "asked the doctor whether he had ever performed any studies related to kratom, and he described an unpublished study from 2020 to 2024 looking at reported deaths in Florida from kratom as compared to a control group of patients that died in Orange County, Florida, who had kratom or mitragynine on board, but whose causes of death were not attributed to those substances." ECF No. 236, Order and Reasons, at 2.

The Court found that Plaintiff "failed" to comply with her obligations under Rule 26(a)(2)(B) to disclose "a complete and accurate written report by Dr. Thundiyil" and that "[g]ood cause" exists to allow a second deposition of Dr. Thundiyil and the disclosure and production of facts or data related to his study. ECF No. 236, at 14–15. The Court set a production deadline of March 3, 2026 and a deadline for completion of the second deposition of Dr. Thundiyil of March 6, 2026. *Id.* Accordingly, Defendants reserve the right to supplement the arguments herein based on the findings from this additional study. This includes challenging Dr. Thundiyil's qualifications as an expert on the pharmacology of kratom or risk profile of Kratom.

In his report, Dr. Thundiyil concludes that the circumstances of Harmony Moller's death "support[] the conclusion that Harmony Moller died from an adverse cardiorespiratory event caused by kratom." Ex. 1, at 10. He asserts "[t]he toxicology results from NMS Labs directly support kratom's causative role" in the death and that the post-mortem blood concentration of mitragynine was "a level sufficient to cause adverse events." *Id.* He also offers the following opinions regarding other substances: (i) that "[t]he toxicology results also returned a positive result for caffeine, a finding expected in individual who ingested caffeinated products such as coffee; (ii) Delta-9 THC and Delta-9 Carboxy THC were "present at low levels" that are "not consistent with a toxic or fatal ingestion"; (iii) diphenhydramine (present at a concentration of 64 ng/ml) and promethazine (present at a concentration of 18 ng/ml) are "not consistent with a fatal or toxic dose" and "consistent with therapeutic dosing," and thus, "unlikely to be a cause or contributor of Harmony Moller's death." *Id.*

He states, "[t]he toxicology results confirm that the cause of death was directly related to kratom." *Id.* His conclusion based on "the historical context, the medical examiner's report, the

heart pathology report, and the toxicology results" is "that kratom caused the adverse cardiorespiratory death of Harmony Moller." *Id.*

Dr. Thundiyil admits there are multiple foundational limitations with his opinion:

- "[I]t's nearly impossible at this point, with the existing science, to have a dose that we know causes harm." Ex. 2, at 147:2–3.
- "We don't know the exact level [of kratom] that causes toxicity." *Id.* at 146:8–9.
- He cannot "back extrapolate" what dose would put a person at risk for fatal arrhythmia, "especially in people who might be taking therapeutic doses of other drugs, like Phenergan and Benadryl in [this] case." *Id.* at 162:11–163:1.
- He acknowledged that there is uncertainty in interpreting post-mortem concentrations of mitragynine and agreed that, if postmortem redistribution occurs, the number "can actually go both ways." *Id.* at 209:11–22.
- He does not know whether the fatal sequence was cardiac failure or respiratory failure: "there's no way to really know for sure which one caused her death." *Id.* at 161:9–16.
- He testified he applied the Bradford Hill criteria but did not do a step-by-step written analysis—he did it "mentally." *Id.* at 233:6–11.

III. ARGUMENT

A. Standard for admissibility.

The district court has considerable discretion to admit or exclude expert testimony under Fed. R. Evid. 702. *See Gen. Elec. Co. v. Joiner*, 522 U.S. 136, 138–39 (1997); *Seatrax, Inc. v. Sonbeck Int'l, Inc.*, 200 F.3d 358, 371 (5th Cir. 2000); *Tajonera v. Black Elk Energy Offshore Operations,* LLC, Civ. A. No. 13-0366, 2016 WL 3180776, at *8 (E.D. La. June 7, 2016) (Brown, J.) (citing authority). The burden rests with the party seeking to present the testimony to show that

the requirements of Rule 702 are met. *Moore v. Ashland Chemical, Inc.*, 151 F.3d 269, 276 (5th Cir. 1998). Rule 702 provides that an expert witness "qualified . . . by knowledge, skill, experience, training, or education may testify in the form of an opinion" when the following requirements are met:

(a) The expert's scientific, technical, or other specialized knowledge will help the trier of fact to understand the evidence or to determine a fact in issue;

(b) The testimony is based on sufficient facts or data;

(c) The testimony is the product of reliable principles and methods; and

(d) The expert has reliably applied the principles and methods to the facts of the case.

Fed. R. Evid. 702. Rule 702 codifies the Supreme Court's opinion in *Daubert,* which charges district courts to act as "gatekeeper" when determining the admissibility of expert testimony. *Tajonera*, 2016 WL 3180776, at *8. The purpose of a *Daubert* motion is "to ensure that only reliable and relevant expert testimony is presented to the jury." *Rushing v. Kansas City S. Ry. Co*., 185 F.3d 496, 506 (5th Cir. 1999) (superseded by rule on other grounds) (citing *Daubert*, 509 U.S. at 590–93).

In the Fifth Circuit, Thundiyil's opinions are inadmissible because they lack a reliable foundation.[1] The Fifth Circuit requires a two-step showing: general causation first (that a

[1] "Plaintiff bears the obligation to disclose a complete and accurate written report by Dr. Thundiyil as described under Rule 26(a)(2)(B)." ECF No. 236, Order and Reasons, at 14–15. The Court found that Plaintiff "failed to do so" and that "[g]ood cause" exists to allow a second deposition of Dr. Thundiyil and the disclosure and production of facts or data ordered by the Court beyond the existing discovery deadline ("Additional Discovery"). *Id.* The Court set a production deadline of March 3, 2026 and a deadline for completion of the second deposition of Dr. Thundiyil of March 6, 2026. *Id.* Accordingly, Defendants reserve the right to supplement the arguments herein based on the findings from the Additional Discovery, including requesting the Dr. Thundiyil be prevented from using his study or related publications to support any of his opinions articulated in the Additional Discovery that could not be raised here.

substance is capable of causing the injury under relevant exposure conditions), and then specific causation (that it did so in this case). *Knight*, 482 F.3d at 351. General causation, at minimum, requires scientific knowledge of a harmful level of exposure. *Id.*; *Allen v. Pa. Eng'g Corp.*, 102 F.3d 194, 199 (5th Cir. 1996).

In his deposition, Dr. Thundiyil conceded that "we don't know the exact level [of kratom] that causes toxicity" and it is "nearly impossible . . . with the existing science, to have a dose that we know causes harm." Thundiyil Dep., at 146:8–9, 147:3–5. He also admitted that it is not possible to "back extrapolate" what dose would produce the alleged injury in this case. *Id.* 162:11–14.

Despite those admissions, Dr. Thundiyil's purported conclusion is that Harmony Moller's post-mortem mitragynine levels show that kratom caused her death. Thundiyil Report, at 10. That is not a reliable scientific inference—it is a leap across an admitted methodological gap. The Fifth Circuit does not allow experts to bridge that gap with labels or by claiming "it is so." *See, e.g.*, *Knight*, 482 F.3d 347, 355 (5th Cir. 2007) (explaining that the court must ensure reliability "at each and every step" and exclude opinions with analytical gaps); *Viterbo v. Dow Chem. Co.*, 826 F.2d 420, 424 (5th Cir. 1987) (explaining that if an opinion is "fundamentally unsupported," it offers no assistance to the jury).

Because Plaintiff cannot carry her Rule 702 burden for general causation or specific causation, the Court should exclude Dr. Thundiyil's testimony in full.

B. Dr. Thundiyil cannot establish general causation and the Fifth Circuit does not permit him to guess.

Dr. Thundiyil admits that with respect to kratom, no toxic threshold is known, no fatal dose range exists, no reliable extrapolation can be performed, and the fatal mechanism itself cannot be determined. Under controlling Fifth Circuit precedent, that is not a dispute over the weight of

the evidence—that is, not a matter for the jury to resolve by deciding how persuasive competing scientific opinions may be. It is an absence of admissible general causation. Dr. Thundiyil admits there is no known harmful level. That ends the inquiry.

The Fifth Circuit requires that before an expert may tell a jury a substance caused a death, he must identify—with reliable science—that the substance is capable of causing that type of death at the levels at issue. The Fifth Circuit requires admissible general-causation evidence before any specific-causation testimony may be admitted. *Johnson v. Arkema, Inc.*, 685 F.3d 453, 469 (5th Cir. 2012) ("First, the district court must determine whether there is general causation. Second, if it concludes that there is admissible general-causation evidence, the district court must determine whether there is admissible specific-causation evidence."); *Knight*, 482 F.3d at 351. At minimum, general causation requires scientific knowledge of a harmful level of exposure. *Johnson*, 685 F.3d at 469 ("Scientific knowledge of the harmful level of exposure to a chemical, plus knowledge that the plaintiff was exposed to such quantities, are minimal facts necessary to sustain the plaintiffs' burden in a toxic tort case."); *Allen*, 102 F.3d at 199 (same); *Braggs v. BP Expl. & Prod., Inc.*, No. 23-30297, 2024 WL 863356, at *3 (5th Cir. Feb. 29, 2024) (affirming exclusion of general causation opinion where expert failed to identify level of exposure capable of causing injuries); *McGill v. BP Expl. & Prod., Inc.*, 830 F. App'x 430, 433 (5th Cir. 2020) (same).

Dr. Thundiyil concedes the very scientific gap the Fifth Circuit says must be bridged before any causation opinion can reach a jury. He cannot identify:

- a toxic threshold,
- a fatal range,
- a reliable dose-response relationship,
- or a mechanism of harm he can apply to this case.[2]

Yet he proposes to tell the jury that the mitragynine level here was "sufficient to cause adverse events" and that kratom caused Ms. Moller's death.

The Fifth Circuit does not allow that leap. If an expert cannot identify the conditions or levels at which a substance can cause the injury, he cannot reliably opine that the exposure in this case was sufficient. *Braggs*, 2024 WL 863356, at *3; *McGill*, 830 F. App'x at 433. His testimony must be excluded because it does not support general causation.

C. Dr. Thundiyil's "Bradford Hill" reference does not cure the general causation gap here.

Dr. Thundiyil's opening expert included less than two-and-a-half pages of substantive explanation for his opinions. *See* Ex. 1, at 8–11. At his deposition, when asked if he "intend[s] to offer any opinions in this case other than the ones in [his] report," Dr. Thundiyil responded "Well, yeah. . ." and proceeded to list various subjects he omitted from his report including "the process by which I developed my opinions, and causation." Ex. 2, at 228:9–16. As to the omitted "process" for determining "whether or not it's even possible for kratom to cause death," Dr. Thundiyil claims he utilized "a well recognized causation process known as the 'Bradford Hill' criteria." *Id*. at 229:14–18.

[2] Dr. Thundiyil cannot explain the mechanism for how kratom may cause death. Rather, he conceded that "we don't know the translation" and cannot "back extrapolate" what dose would produce the alleged adverse effect. Ex. 2, at 162:11–20.

This Court is familiar with the "so-called Bradford Hill factors" since another expert witness proffering excludable opinions also attempted to "describe[] how he relied upon the[m]" in *Kaoui v. BP Expl. & Prod., Inc*., No. CV 17-3313, 2023 WL 330510, at *2 (E.D. La. Jan. 12, 2023) (Vitter, CJ). In *Kaoui*, the Court explained its understanding that "[a]fter evidence demonstrating an association between a particular chemical and a disease has been established, the Bradford Hill criteria are used to determine whether a causal relationship exists." *Id*.; *see also id.*, *2 n. 22 ("The Bradford Hill criteria include: "(1) Temporal relationship, (2) Strength of the association, (3) Dose-response relationship, (4) Replication of the findings, (5) Biological plausibility, (6) Consideration of alternative explanations, (7) Cessation of exposure, (8) Specificity of the association, and (9) Consistency with other knowledge.")

In *Kaoui*, despite his invocation of the Bradford Hill criteria, the plaintiff admitted that its expert—who the Court ultimately excluded under Rule 702—did "not employ a 'traditional Bradford Hill dose-response relationship' in his analysis" because "he 'does not identify a particular dose to a particular chemical.'" *Kaoui*, 2023 WL 330510, at *8. Here, despite invoking the Bradford Hill criteria and reciting them at his deposition, Dr. Thundiyil has failed to show how he employed *any* of the criteria in his analysis—not just the dose-response relationship. *See generally*, Thundiyil Report (providing no analysis of *any* Bradford Hill criteria). Indeed, at his deposition, Dr. Thundiyil admitted that he did not, in fact, do any "step-by-step analysis in writing." Ex. 2, at 233:6–8.

An undocumented, unarticulated "mental" Bradford Hill analysis is not a methodology. It is an assertion. And assertions do not satisfy Rule 702. "'To ensure that the Bradford Hill/weight of the evidence criteria is truly a methodology, rather than a mere conclusion-oriented selection process, there must be a scientific method of weighting that is used and explained.'" *In re*

Paraquat Prods. Liab. Litig., 730 F. Supp. 3d 793, 841 (S.D. Ill. 2024) (quoting *In re Zoloft (Sertraline Hydrochloride) Prods. Liab. Litig.*, 858 F.3d 787, 796 (3d Cir. 2017) (cleaned up). "With these considerations in mind, an expert who relies on a weight of the evidence review based on Bradford Hill framework must, at a minimum, 'explain how conclusions are drawn for each Bradford Hill criterion and how the criteria are weighed relative to one another.'" *Id*. (quoting *Zoloft*, 858 F.3d at 796) (cleaned up).

Dr. Thundiyil attempts to invoke Bradford Hill. But he admitted it was "just something [he] did mentally in [his] head." Ex. 2, at 233:6–11. Rule 702 does not accept mental checklists. Rule 702 demands a transparent, testable reasoning process. *Moore*, 151 F.3d at 276 ("The expert's assurances that he has utilized generally accepted scientific methodology is insufficient."); *see Konrick v. Exxon Mobil Corp.*, No. CV 14-524, 2016 WL 439361, at *11 (E.D. La. Feb. 4, 2016) (granting *Daubert* motion and finding that an expert witness's opinion was "conclusory *ipse dixit*" where an expert's report did not explain her methodologies or how she applied them to the scientific literature). Courts evaluate what the expert actually did, not what he says he did. *Moore*, 151 F.3d at 276. The court must ensure reliability "at each and every step" and exclude opinions with analytical gaps. *Knight*, 482 F.3d at 355. The mental Bradford Hill analysis does not save his testimony.

What Dr. Thundiyil offers is not a structured epidemiologic application of Bradford Hill, but a retrospective justification for a conclusion he had already reached. Without identifying which factors he relied upon, how he weighed them, what data supported each factor, or how the criteria bridge the admitted dose and mechanism gaps in this case, his invocation of Bradford Hill is just an assertion. Rule 702 requires more than the label of a methodology—it requires its disciplined application. Because that showing is absent here, the Court should exclude the testimony.

D. Dr. Thundiyil's specific causation opinions are entirely unreliable

1. Dr. Thundiyil cannot establish specific causation because he cannot identify a harmful dose.

Dr. Thundiyil claims to have applied a "differential diagnosis" to assess specific causation. Ex. 2, at 230:6–11. Under Fifth Circuit precedent, his failure on general causation means his differential diagnosis cannot be admissible. *Johnson*, 685 F.3d at 469 ("before courts can admit an expert's differential diagnosis, which, by its nature, only addresses the issue of specific causation, the expert must first demonstrate that the chemical at issue is actually capable of harming individuals in the general population, thereby satisfying the general causation standard").

Because Dr. Thundiyil cannot identify a harmful dose of kratom, he cannot determine whether Moller consumed enough kratom to cause her death. "[N]early every chemical on Earth may be toxic or even fatal at a certain level of exposure," so "[c]ausation experts determine not only *whether* a chemical is capable of causing certain health effects, but *at what level* of exposure do those health affects appear." *Kaoui v. BP Expl. & Prod., Inc*., No. CV 17-3313, 2023 WL 330510, at *6 (E.D. La. Jan. 12, 2023) (italics original). "Thus, causation experts determine not only whether a chemical is capable of causing certain health effects, but at *what* level of exposure do those health affects appear." *Id*. (italics original). Dr. Thundiyil testified that after his general-causation step, he applied a differential diagnosis and asked whether "the exposure [was] high enough to cause" Moller's death. Ex. 2, at 230:6–22. But he also admitted the science does not allow him to identify a dose of kratom or level "we know causes harm." *Id.* at 146:5–10; 147:1–7. Under *Johnson*, that concession alone is dispositive. Without evidence of a harmful level of exposure, specific causation cannot stand.

2. Dr. Thundiyil's specific causation opinion is based on unjustified assumptions

Even where general causation is established, the Fifth Circuit excludes opinions where the reasoning amounts to assuming causation simply because the event happened after exposure, rather than reliable science. *Black v. Food Lion, Inc.*, 171 F.3d 308, 313–15 (5th Cir. 1999). Where assumptions substitute for facts—such as, for example, exposure, as here—causation devolves into "mere guesswork" and is inadmissible. *Seaman v. Seacor Marine L.L.C.*, 326 F. App'x 721, 727–28 (5th Cir. 2009) (affirming exclusion where assumptions about exposure levels rendered causation opinion "mere guesswork" and failed to establish either general or specific causation).

Even apart from his foundational failure on general causation, Dr. Thundiyil's specific-causation opinion rests on inputs that cannot be determined. He acknowledged that redistribution of kratom in the deceased's body following death can cause mitragynine levels to appear higher or lower than they were before death—"it can actually go both ways." Ex. 2, at 209:11–22. Yet he treats the post-mortem concentration as if it answers the key toxicology questions of sufficiency and cause. And he admitted he cannot say for certain whether the death sequence was respiratory-first or cardiac-first—"there's no way to really know for sure which one caused her death." *Id.* at 161:9–16. These are not minor uncertainties. They go to the core of the causation analysis. The Fifth Circuit repeatedly warns that courts must exclude medical testimony where the expert cannot reliably connect the scientific literature to the facts of the case. *Knight*, 482 F.3d at 355. That is precisely the "analytical gap" Daubert forbids.

Black is particularly instructive. There, the Fifth Circuit rejected the precise type of testimony Thundiyil wants to give here—the inference that because symptoms (or an injury) followed an exposure, the exposure must have caused the injury. *Black*, at 171 F.3d at 313. In *Black*, the plaintiff slipped on mayonnaise at a grocery store and claimed the fall caused

fibromyalgia. *Id.* at 309. The expert had reviewed the plaintiff's medical history, ruled out other potential causes, and hypothesized that the fall must be the cause because nothing else was left. *Id.* at 310. The Fifth Circuit reversed, observing that the expert's opinion lacked the "intellectual rigor" *Daubert* requires and rested on conjecture rather than scientifically validated reasoning. *Id.* at 313. The theory had not been verified by testing, had not been peer-reviewed, had no known potential rate of error, and fibromyalgia's etiology and triggering factors were unknown to medical science, making the expert's causation opinion an unsupported leap. *Id.*

Dr. Thundiyil's opinions should be excluded for the same reasons. *See id.* Dr. Thundiyil's causation theory reduces to this: kratom was present, death followed, therefore kratom caused the death. Yet he admits there is no scientific basis establishing that the detected level is capable of causing fatal harm. Ex. 2, at 146:5–10, 147:1–7, 230:6–22.

Dr. Thundiyil cannot identify a harmful level of exposure and admits the toxicology cannot reliably show what dose was present at death. Without reliable general causation or a sound scientific foundation for his differential diagnosis, his specific-causation opinion is inadmissible.

3. Dr. Thundiyil's "rule-out" opinions are inadmissible because they are not based on a reliable method that accounts for alternative causes.

Dr. Thundiyil's "differential diagnosis" further fails because he rejects other substances as potential causes because they were not present at a "fatal or toxic" level, yet admits he cannot identify any known "fatal or toxic" level for kratom—an inconsistency that makes his causation analysis scientifically unreliable. An expert cannot simply discard alternative causes without explanation and then ask the jury to fill the gap. That analytical leap is a Rule 702 reliability failure—not a credibility question. *Knight*, 482 F.3d at 355 ("Of the over fifty studies relied upon by Dr. Levy, none gave an adequate basis for the opinion that the types of chemicals appellants were exposed to can cause their particular injuries in the general population").

The Fifth Circuit has affirmed exclusion where the expert's approach failed to address other potential causes or showed an analytical gap between relied-on facts and conclusions. *Viterbo*, 826 F.2d at 424 (an expert's "admission that the [plaintiff's] symptoms could have a number of causes" means his opinion that one particular substance "caused [plaintiff's] illness is unfounded"); *McGill*, 830 F. App'x at 433 (affirming exclusion of expert where "there is a notable analytical gap between the facts he relies on and the conclusions he reaches"). Dr. Thundiyil attempts to rule out other substances—THC, diphenhydramine, and promethazine—as "not consistent with a fatal or toxic dose," and to dismiss alternative medical explanations. Yet he simultaneously admits that he cannot identify a "fatal or toxic" threshold for kratom. Ex. 2, at 146:5–10; 147:1–7; 162:11–13. That asymmetry is fatal. He applies a quantitative standard to exclude alternatives while conceding no such scientifically established threshold exists for kratom. In effect, kratom is deemed "sufficient" without reliable dose-response science, while competing causes are rejected under a standard he cannot consistently apply. That is not differential diagnosis—it is result-driven reasoning. That is the sort of analytical gap *Viterbo* and *McGill* treat as inadmissible. *Viterbo*, 826 F.2d at 424 (where plaintiff's "symptoms could have numerous causes and, without support" the expert "simply picks the cause that is most advantageous to [plaintiff's] claim," the expert's testimony is properly excluded); *McGill*, 830 F. App'x at 433 (affirming exclusion of expert testimony on causation because of its unreliability, its lack of evidence of a link between the chemical that allegedly caused the harm and the precise illness, and because there was no evidence of the level of exposure).

Because Dr. Thundiyil's rule-out opinions are not the product of a reliably applied methodology and are inseparable from the unreliable causation theory, they should be excluded along with the rest of Dr. Thundiyil's causation testimony.

IV. CONCLUSION

Defendants respectfully request that the Court exclude Dr. Josef Thundiyil's opinions in their entirety, including any opinions that: (1) kratom caused Harmony Moller's death; (2) the reported mitragynine concentration was "sufficient" to cause adverse events or death; and (3) other potential causes or contributors were reliably ruled out. Defendants request a hearing for argument on this motion prior to trial in this case.

Dated this 2nd day of March, 2026.

Respectfully Submitted

By: */s/ Gwendolyn C. Payton*

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CERTIFICATE OF SERVICE

This is to certify that on the 2nd day of March, 2026, a copy of the foregoing document was filed with the Clerk of Court and served on all counsel of record via the Court's ECF System.

/s/ Gwendolyn C. Payton
Gwendolyn C. Payton